K711 — CKM CLOSE-OUT: V_cb forward-derives to 0.044 at STRUCTURAL tier via the projection-truncation mechanism (BANKED S-tier). The sub-percent 0.041 + 36/869 are FITS (REJECTED). And the observed V_cb isn't even pinned to sub-percent — a 5% inclusive/exclusive puzzle — so structural is the honest tier, not a shortfall.

Keeper | 2026-07-16 evening | Auditing the Grace (0.044 forward-derives) + Lyra (sub-percent doesn't) + Elie (J rides on V_cb) convergence. Resolves the Grace/Lyra "tension" (two radius provenances), banks V_cb at structural tier, rejects the sub-percent fit, and closes the CKM sector at honest mixed tiers.

THE RESOLUTION — two provenances for one radius; the projection one derives

Grace and Lyra were not in conflict — they tested DIFFERENT origins for the 23-radius: - Lyra (excitation-ladder provenance, toy_4620): FAILS. r_μ=1/2 wants α=4.5, r_τ=$\sqrt{6}/3$ wants α=1.25 — inconsistent Bergman weights for one sector → the radii would be hand-set → fit. Correctly rejected. - Grace (projection-fraction provenance): WORKS. r_τ = $\sqrt{(C_2/(C_2+N\_c))}$ = $\sqrt{(2/3)}$ = 0.8165 (= Casey's 3D→2D hemispherical-lens fraction = Keeper's $\sqrt{(C_2)}$/N_c, same number) → V_cb = 0.044, forward, target-innocent. Casey's projection insight picked the correct origin. The radius is the holographic projection fraction, NOT the excitation-ladder radius. That is why Lyra's test failed (wrong provenance) and Grace's works.

THE MECHANISM — clean, and it IS Casey's truncation + the banked y_t=1

- The ×3/2 = N_c/rank refraction pushes the up-sector 23-radius to 0.8165 × 1.5 = 1.225 > 1 → PAST the domain boundary (|z|<1 interior). The up-sector 23-mode projects outside the aperture → vanishes. This is literally Casey's "projection truncated at the external boundary / hemispherical lens."
- Up vanishing = the top quark saturating the boundary (y_t=1, banked: m_t=(1−α)v/$\sqrt{2}$). So the V_cb mechanism reuses a banked result — not a new assumption.
- V_cb = the down sector alone, at the projection radius $\sqrt{(2/3)}$, through the genus-5 F379 kernel → 0.044.
- DIRECTION derived (Lyra F384): cos ψ = n_C/$\sqrt{(n\_C^2+N\_c^2)}$ = 5/$\sqrt{34}$ (tau floor-position $\hat{\rho}$ vs excitation axis $\hat{e}_1$, bare integers).

THE TIER — STRUCTURAL, and that's the HONEST tier (not a shortfall)

- V_cb = 0.044. Observed: **|V_cb|_incl = 0.0420 (4.8% below), |V_cb|_excl ≈ 0.0390 (~13% below), PDG avg ≈ 0.0405 (~8% below).**
- KEY (web, load-bearing): the observed V_cb is NOT pinned to sub-percent — the inclusive/exclusive "V_cb puzzle" is a ~5% (2.0–2.6σ) discrepancy, unresolved for 20 years ("we do not understand why," PDG). So demanding sub-percent from BST was chasing a precision the DATA does not have. BST's 0.044 sits ~5% above the higher (inclusive) determination — an honest structural prediction, and it lands closest to the inclusive value.
- VERDICT: V_cb BANKS at STRUCTURAL tier (S), 0.044. Mechanism derived (down-alone via refraction-truncation = y_t=1; projection-fraction radius $\sqrt{(2/3)}$, target-innocent form), direction derived (5/$\sqrt{34}$). This is consistent with the Two-Tier hypothesis (mixing = Tier-2 structural ~%).

WHAT'S REJECTED (fits — do NOT bank)

- The sub-percent 0.041: needs r_τ = 0.8207 (only 0.5% off $\sqrt{(2/3)}$), but the genus-5 power swings V_cb ~8% per 0.5% in radius → any radius near 0.82 fits any V_cb. This is Grace's K588 fish ("cleanness in the wrong place"), correctly refused again.
- 36/869 = $C_2^2/(11\cdot 79)$: the 11 and 79 have no geometric source (Grace + Cal + Keeper K708 all agree). Asserted form, not a derivation. OUT.

RESIDUAL RIGOR (the one upgrade path for V_cb: structural → derived)

The projection-fraction radius $\sqrt{(2/3)}$ is target-innocent as a FORM and physically grounded (3D→2D RMS = $\sqrt{(C_2/(C_2+N\_c))}$; the $/\mathbb{Z}_2$ Shilov hemisphere; the refraction-truncation = y_t=1). But "gen-3 23-radius = the 3D→2D projection fraction" is a physically-motivated IDENTIFICATION, not yet a theorem. Turning Casey's hemispherical-lens truncation into a derivation of the projection radius is the one path to upgrade V_cb from structural-identified to derived. Until then: structural, banked, honest.

CP / J (Elie) — rides on V_cb, so structural

J = (Cabibbo)(V_cb)(V_ub)(sin δ), linear in V_cb; phase enters via the FK "1−r_i r_j·ω" twist (not a rephasing-removable bare $\mathbb{Z}_3$). So J ≈ 3×10^{-5} is a STRUCTURAL prediction inheriting V_cb's tier. Mechanism sound.

V_ub — FLAG (check before declaring CKM fully closed)

|V_ub/V_cb| ~ $\sqrt{(m\_u/m\_c)}$ ≈ 0.045 → V_ub ≈ 0.044×0.045 ≈ 0.0020 vs observed 0.0037 — ~factor 1.8 (the known Fritzsch V_ub weakness). V_ub is the un-audited third angle; likely structural-or-worse. Confirm its tier before the CKM sector is called closed.

CKM SECTOR STATUS (honest, mixed tiers)

- Mechanism: PROVED (one Higgs → rank-1 → Gatto), target-innocent, falsifiable (2nd Higgs breaks it). CONDITIONAL PASS.
- V_us: leading-order Gatto, ~0.4%, pending Cal's F506 circularity trace. Near-closed.
- V_cb: BANKED structural 0.044 (mechanism derived, radius target-innocent; sub-percent rejected).
- δ_CKM / J: structural, rides on V_cb.
- V_ub: un-audited, likely structural-or-miss — CHECK. CKM is mechanism-closed with V_us + V_cb derived at honest tiers. The big remaining block is now PMNS (neutrino/Majorana).

Lead (Lyra, flagged not claimed)

up/down 23 mass-ratio ratio = 0.574 ≈ $1/\sqrt{N\_c}$ = 0.577. Possible refraction signature or coincidence. Pin or drop — don't bank.

Sources (V_cb puzzle): PDG 2024 Vcb/Vub review · inclusive/exclusive tension

— Keeper K711, 2026-07-16. V_cb BANKS structural 0.044 via projection-truncation (up refracts past boundary = y_t=1; radius $\sqrt{(2/3)}$ = Casey's hemispherical lens, target-innocent). Sub-percent 0.041 + 36/869 REJECTED as fits (genus-5 sensitivity + unsourced 11,79). Observed V_cb has a 5% inclusive/exclusive puzzle → structural is the honest tier. Residual: make the projection-radius a theorem (structural→derived). V_ub un-audited. CKM mechanism-closed; PMNS is the next block. See [[Keeper_K710...]], [[Keeper_K708...]].